

PGCF+AGCF: A Multi-Modal Audio-Visual Deepfake Detection Framework via Physiology- and Audio-Guided Consistency with Hard AND-Gate Late Fusion

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Abstract—Video-only deepfake detectors are structurally blind to audio-driven forgeries such as lip-synced voice cloning, where the visual track remains pristine while only the speech is synthetic. We extend our previously reported Physiology-Guided Consistency Framework (PGCF) – a three-stream video detector fusing spatial, frequency, and remote photoplethysmography (rPPG) cues – with a newly designed audio companion network, the Audio-Guided Consistency Framework (AGCF). AGCF mirrors PGCF’s temporal backbone (Temporal Gradient Fusion, 4-head self-attention, BiLSTM(256), Classifier(512→2)) so that the two independently trained branches remain directly comparable; its dual-stream front end pairs a log-mel spectral CNN (the audio analogue of PGCF’s DCT frequency stream) with a raw-waveform 1D-CNN (the analogue of PGCF’s rPPG stream). The two branches are combined at decision level, and we contrast a continuous Soft (probability-product) fusion rule against a conservative Hard AND-gate that flags a clip as fake only when *both* modalities independently agree. On an identity-disjoint, held-out evaluation of the FakeAVCeleb corpus, PGCF reaches an AUC of 0.999 (98.5% accuracy), AGCF reaches an AUC of 0.884 (78.8% accuracy), Soft fusion reaches an AUC of 0.979 (93.4% accuracy), and the Hard AND-gate achieves 100% precision (0% false-positive rate on real videos) at 83.8% accuracy. We further document a data-integrity audit that uncovered and corrected two evaluation flaws – identity leakage across the train/validation split and a filename-collision-induced Cartesian merge – and show that the naive (uncorrected) evaluation had inflated apparent performance, underscoring the importance of identity-disjoint protocols for audio-visual deepfake benchmarking. Robustness testing under additive Gaussian noise, ROC/PR analysis, calibration curves, and t-SNE feature-space visualization are reported for all four configurations.

Index Terms—deepfake detection, audio spoofing detection, multi-modal fusion, remote photoplethysmography, cross-modal consistency, late fusion, identity-disjoint evaluation, CBAM, BiLSTM, temporal self-attention

I. INTRODUCTION

Face manipulation technologies, colloquially termed deepfakes, have reached a level of visual fidelity that renders

pixel-level inspection by human observers unreliable. Generative Adversarial Networks (GANs), autoencoder-based face-swapping, and neural rendering pipelines can produce convincing videos, with direct implications for misinformation, identity fraud, and political manipulation [1].

In our earlier work we introduced the Physiology-Guided Consistency Framework (PGCF), a three-stream video detector that fuses spatial (ResNet50+CBAM), frequency (DCT), and physiological (CHROM rPPG) evidence under a learned cross-modal consistency gate, achieving an AUC of 0.824 on a balanced Celeb-DF-v2 validation set. While effective against classical face-swap and re-enactment forgeries, PGCF is a purely visual pipeline and is therefore structurally blind to *audio-driven* deepfakes – most notably lip-sync attacks such as Wav2Lip [12], in which the visual frames remain entirely authentic and only the speech track is synthesized or voice-cloned. Against such attacks, a video-only detector produces severe false negatives regardless of how sophisticated its visual reasoning is, because the manipulation simply does not exist in the modality it inspects.

This paper closes that gap. We introduce the **Audio-Guided Consistency Framework (AGCF)**, an audio companion network deliberately architected as a stream-for-stream mirror of PGCF: a spectral CNN plays the role of PGCF’s frequency stream, a raw-waveform 1D-CNN plays the role of PGCF’s rPPG stream, and the fused 512-d representation is pushed through an *identical* Temporal Gradient Fusion → 4-head self-attention → BiLSTM(256) → Classifier(512→2) tail, trained with matched hyperparameters. This deliberate architectural symmetry is what makes the two branches’ output probabilities directly comparable and safe to fuse at the decision level without any re-calibration heuristics.

We then study two decision-level fusion strategies – a continuous Soft (probability-product) rule and a conservative Hard AND-gate that requires unanimous agreement – and

show that the AND-gate drives the false-positive rate on real videos to zero while PGCF alone continues to catch the visual attacks and AGCF alone continues to catch the audio-only attacks that PGCF cannot see.

Finally, in the course of building the identity-disjoint evaluation protocol for AGCF on FakeAVCeleb [9], we discovered and fixed two data-integrity bugs – subject-level leakage between train and validation splits, and a filename-collision Cartesian-merge bug that silently duplicated “Real” predictions during audio/video score merging. We report both the naive (buggy) and corrected metrics side by side, because the gap between them is itself an instructive result for the audio-visual deepfake benchmarking community.

Main contributions:

- **AGCF**, a dual-stream (spectral + vocal) audio deepfake detector whose temporal backbone is architecturally matched to PGCF’s, enabling apples-to-apples decision-level fusion.
- A formally defined **Hard AND-gate fusion** rule, contrasted against Soft (product) fusion, with a full truth-table analysis of the resulting precision/recall trade-off.
- An **evaluation-integrity audit** of the FakeAVCeleb training/validation pipeline that identifies and corrects identity leakage (130 overlapping subjects) and a Cartesian-merge metric-inflation bug, with before/after comparison tables.
- A **comprehensive multi-modal evaluation suite**: ROC/PR curves, confusion matrices, calibration curves, t-SNE feature-space visualization, and Gaussian-noise robustness analysis for PGCF, AGCF, and both fusion strategies.
- A documented, code-level pipeline for **cross-dataset generalization** testing on LAV-DF [10], reported here as an engineering contribution with results left as ongoing/future work rather than fabricated.

II. RELATED WORK

A. Spatial and Frequency Deepfake Detection

FaceForensics++ [2] established a large-scale benchmark showing that CNNs trained on RGB frames achieve high in-distribution performance but degrade under compression. Frank et al. [3] demonstrated that GAN-generated images exhibit characteristic frequency artifacts, motivating DCT-based features – the design basis of PGCF’s frequency stream and, by extension, AGCF’s spectral stream.

B. Physiological Signal-Based Detection

Li and Lyu [4] first exploited warping artifacts as a deepfake indicator. FakeCatcher [5] demonstrated real-time detection using blood-volume-pulse maps but remains brittle to illumination variation and motion blur – the CHROM-based rPPG stream in PGCF, and its audio analogue (the raw-waveform vocal stream in AGCF), inherit this sensitivity, which motivates our robustness analysis under additive noise (Section V-F).

C. Audio Spoofing and Synthetic Speech Detection

A parallel literature addresses *audio-only* spoofing. AASIST [11] models artefacts across heterogeneous spectral and temporal intervals using graph attention networks and represents the current state of the art on ASVspoof-style benchmarks. Unlike AASIST’s graph-based backbone, AGCF intentionally reuses a convolutional-recurrent-attention backbone identical to PGCF’s, trading a small amount of raw audio-only accuracy for direct architectural comparability across modalities – our design goal is a matched pair of detectors rather than a maximally optimized standalone audio spoofing system.

D. Multi-Modal and Temporal Approaches

Recurrent and attention-based architectures have been applied to temporal inconsistency modeling [6]. FakeAVCeleb [9] and LAV-DF [10] were introduced specifically to support audio-visual (rather than visual-only) deepfake research, and are the datasets used for AGCF training/validation and cross-dataset generalization testing, respectively, in this work. Our fusion strategy is closest to late-fusion multi-modal detectors but distinguishes itself by (i) enforcing architectural symmetry between the unimodal branches and (ii) formally comparing a probabilistic (Soft) fusion rule against a logical (Hard AND) rule rather than relying on a single learned fusion head.

III. METHODOLOGY

A. System Overview

Given a video clip, two independently trained branches each produce a probability of the clip being fake: $P(\text{fake} \mid \text{video})$ from PGCF and $P(\text{fake} \mid \text{audio})$ from AGCF. Both branches are trained *independently* – PGCF is left completely untouched from our earlier work – and are combined only at inference time via a late-fusion rule (Section III-D). Figure ?? (reproduced from our earlier work) and Figure ?? summarize the two branches; observe the deliberate stream-for-stream and stage-for-stage correspondence between them.

1) *PGCF Recap*: PGCF processes $T = 16$ sampled frames through three parallel streams: (i) a Spatial stream (ResNet50 backbone with a CBAM [8] attention block) producing a 256-d embedding per frame; (ii) a Frequency stream (3-layer 2D-CNN over a 4-channel RGB+DCT-energy input) producing a 256-d embedding per frame; and (iii) an rPPG stream (1D-CNN over a CHROM [7] blood-volume-pulse signal) producing a 128-d embedding. The Physiology-Guided Cross-Modal Consistency (PGCMC) module computes an rPPG spectral-entropy confidence c and a pairwise cosine-similarity consistency score s , and uses both to gate the fusion of the Spatial and Frequency streams into a 512-d representation. This representation is passed through Temporal Gradient Fusion (TGF), a 4-head Transformer self-attention layer, and a BiLSTM(256), before a final Linear(512→2) classifier. Full architectural and training detail is given in our earlier report; PGCF is not modified in this work.

TABLE I: Stream-for-Stream Correspondence Between PGCF and AGCF

Role	PGCF (Video)	AGCF (Audio)
Frequency-domain evidence	DCT energy map $\rightarrow$ 3-layer 2D-CNN $\rightarrow$ 256-d	Log-mel spectrogram (64 mel bins) $\rightarrow$ 3-layer 2D-CNN (4 $\rightarrow$ 32 $\rightarrow$ 64 ch) $\rightarrow$ 256-d
Fine-grained / physiological evidence	CHROM rPPG signal $\rightarrow$ 1D-CNN $\rightarrow$ 128-d	Raw waveform (16,000 samples) $\rightarrow$ 1D-CNN (1 $\rightarrow$ 32 $\rightarrow$ 64 ch) $\rightarrow$ 256-d
Fusion	PGCMC gated concatenation $\rightarrow$ 512-d	Direct concatenation $\rightarrow$ 512-d
Temporal pipeline	TGF $\rightarrow$ 4-head Self-Attn $\rightarrow$ BiLSTM(256)	identical: TGF $\rightarrow$ 4-head Self-Attn $\rightarrow$ BiLSTM(256)
Classifier	Linear(512 $\rightarrow$ 2)	identical: Linear(512 $\rightarrow$ 2)

B. AGCF: Audio-Guided Consistency Framework

AGCF is architected so that every design decision in PGCF has a direct audio counterpart, summarized in Table I.

1) *Spectral Stream*: Audio is resampled to 16 kHz (Librosa) and converted to a log-mel spectrogram with 64 mel bins across 100 temporal frames, producing a (1, 64, 100) tensor per chunk. This is processed by a 3-layer 2D-CNN with the same 4 $\rightarrow$ 32 $\rightarrow$ 64 channel progression used in PGCF’s frequency stream, followed by pooling and a projection to a 256-d embedding. This stream targets unnatural spectral energy distributions and phase irregularities characteristic of vocoder- and TTS-generated speech – the audio analogue of the compression/generative artifacts that PGCF’s DCT stream targets in the visual domain.

2) *Vocal Stream*: The raw 1-second waveform (16,000 samples) is processed directly in the time domain by a 1D-CNN with a 1 $\rightarrow$ 32 $\rightarrow$ 64 channel progression, mirroring the design of PGCF’s rPPG 1D-CNN. The resulting embedding is projected to 256-d for symmetry with the spectral stream. This stream captures fine-grained vocal-tract and micro-acoustic detail – pitch-period irregularities, robotic timbre, and phase discontinuities – that are lost during spectrogram conversion, playing the same structural role that the physiological rPPG signal plays for PGCF: a signal that is difficult for a generative model to reproduce faithfully because it is not what the generator was optimized to fool a listener’s ear on.

3) *Feature Fusion and Temporal Modeling*: The 256-d spectral and 256-d vocal embeddings are concatenated into a 512-d multi-domain representation per audio chunk. Audio is windowed into the same $T = 16$ chunks as the video clip so both branches consume matched clip durations – not required for decision-level fusion, but it keeps the two networks directly comparable and leaves the door open for feature-level fusion in future work. The chunk sequence is then passed through the identical three-stage temporal pipeline used by PGCF:

- 1) **Temporal Gradient Fusion (TGF)**: computes $\text{Feature}_t - \text{Feature}_{t-1}$ to explicitly expose unnatural

TABLE II: Hard AND-Gate Fusion Truth Table

PGCF says	AGCF says	Final verdict
Real	Real	Real
Real	Fake	Real
Fake	Real	Real
Fake	Fake	Fake

frame-to-frame (chunk-to-chunk) transitions such as voice jitter or unnatural phoneme pacing.

- 2) **Temporal Self-Attention**: a 4-head Transformer encoder layer that learns which parts of the clip are most decision-relevant (e.g., a single mispronounced or artificially smoothed word).
- 3) **BiLSTM(256)**: processes the sequence forward and backward to capture sequential pacing and context.

The temporal features are average-pooled into a single 512-d clip-level representation and passed to a final Linear(512 $\rightarrow$ 2) classifier producing Real/Fake logits, trained with the same cross-entropy objective and optimizer configuration as PGCF (Section IV-C).

C. Late Fusion: Soft vs. Hard AND-Gate

Both branches are trained independently, which keeps PGCF completely unmodified. At inference, each branch outputs an independent probability of forgery, and we evaluate two fusion rules.

1) Soft Fusion (Product Rule):

$$P_{\text{fused}} = P(\text{fake} \mid \text{video}) \times P(\text{fake} \mid \text{audio}) \quad (1)$$

This produces a continuous, re-thresholdable confidence score. Because both branches must independently assign high probability for the product to remain high, this rule already suppresses false positives relative to either unimodal branch, while remaining a soft (probabilistic) score suitable for ROC/PR analysis.

2) *Hard Fusion (AND-Gate Logic)*: Let $\hat{y}_v = \mathbb{I}[P(\text{fake} \mid \text{video}) > \tau_v]$ and $\hat{y}_a = \mathbb{I}[P(\text{fake} \mid \text{audio}) > \tau_a]$ denote the binarized PGCF and AGCF decisions at their respective operating thresholds ($\tau_v = \tau_a = 0.5$ throughout this work). The fused verdict is

$$\hat{y}_{\text{AND}} = \begin{cases} \text{FAKE} & \text{if } \hat{y}_v = 1 \text{ AND } \hat{y}_a = 1 \\ \text{REAL} & \text{otherwise} \end{cases} \quad (2)$$

Table II gives the full truth table. Because the AND-gate is a strict binary consensus mechanism, it produces no continuous score and is therefore excluded from ROC/PR/AUC analysis (which require a rankable score) but is fully reportable via a confusion matrix and point-estimate accuracy/precision/recall/F1.

The AND-gate is a deliberately conservative design choice: it trades recall for precision, on the premise that a false accusation of forgery (a false positive on a real video) is often more costly in deployment than a missed detection, especially for single-modality attacks that a human reviewer or an upstream unimodal filter can still catch.

IV. DATASETS AND EVALUATION PROTOCOL

A. Datasets

Celeb-DF-v2 was used exclusively to train and validate the visual-only PGCF baseline in our earlier work, providing a strong foundation of high-quality visual manipulations for spatial artifact learning.

FakeAVCeleb [9] is used for training and validating AGCF and for evaluating the final multi-modal fusion framework in this work. It contains authentic videos alongside multi-modal forgeries (e.g., FaceSwap, Wav2Lip, DeepFaceLab) spanning multiple ethnicities. For direct pairing with AGCF under a matched evaluation protocol, PGCF is evaluated on the same FakeAVCeleb held-out partitions described below (rather than on Celeb-DF-v2); these numbers should therefore be read as same-corpus, paired video/audio performance rather than as a replacement for the original Celeb-DF-v2 baseline result (AUC 0.824, reported in our earlier work).

LAV-DF [10] is used for cross-dataset generalization testing (Section VI-C). We built an auto-resuming extraction pipeline to handle the 23.1 GB archive and a filtration script that isolates a balanced 150-real/150-fake subset, together with a zero-shot cross-evaluation script. This pipeline is a completed engineering contribution of this work; the corresponding zero-shot numerical results were not finalized in time for this submission and are reported as ongoing work (Section VII) rather than estimated or fabricated here.

B. Evaluation Integrity: Identity-Disjoint Splitting

During a pipeline audit of the AGCF training loop on FakeAVCeleb, we identified and corrected two flaws that materially affect reported metrics.

1) *Identity-Level Leakage*: The initial 80/20 split used random file-level shuffling, which left 130 overlapping subject identities between the training and validation partitions. Because multiple clips of the same speaker appeared on both sides of the split, the model could partially memorize speaker-specific voice characteristics rather than learning generalizable deepfake artifacts. We re-engineered the data loader to group all clips by subject identity (e.g., `id00078`) and split at the *subject* level, guaranteeing zero identity overlap and forcing genuine zero-shot evaluation on unseen speakers.

2) *Cartesian-Merge Metric Inflation*: Real videos in FakeAVCeleb frequently share generic filenames (e.g., `00005.mp4` recurs across dozens of subject folders). Merging audio and video prediction tables using filename alone as the join key silently produced a Cartesian-product explosion that duplicated “Real” predictions and artificially inflated accuracy. We corrected this by constructing globally unique keys that prepend the parent subject directory to the filename (e.g., `id00078_00005.mp4`).

3) *Naive vs. Corrected Metrics*: Table III contrasts the larger, uncorrected evaluation partition (372 clips, identity-overlapping, class-imbalanced at 272 real / 100 fake) against the corrected, identity-disjoint held-out set (198 clips, near-balanced at 100 real / 98 fake) used as the primary evaluation

TABLE III: Effect of Identity-Disjoint Correction on Reported Metrics

Model	Split	Acc.	AUC	F1
AGCF (Audio)	Naive (n=372)	0.892	0.923	0.802
	Identity-disjoint (n=198)	0.788	0.884	0.761
PGCF (Video)	Naive (n=372)	0.960	0.995	0.930
	Identity-disjoint (n=198)	0.985	0.999	0.985
Soft Fusion	Naive (n=372)	0.976	0.985	0.954
	Identity-disjoint (n=198)	0.934	0.979	0.929

TABLE IV: Comprehensive Metrics – Identity-Disjoint Evaluation (n=198)

Model	Acc.	Prec.	Rec.	F1	AUC
PGCF (Video)	0.985	0.980	0.990	0.985	0.999
AGCF (Audio)	0.788	0.859	0.684	0.761	0.884
Late Fusion (Soft)	0.934	1.000	0.867	0.929	0.979
Late Fusion (Hard AND)	0.838	1.000	0.673	0.805	–

throughout the remainder of this paper. All results in Section V are reported on the *corrected, identity-disjoint* split unless explicitly marked otherwise.

The audio branch shows the expected direction of the leakage effect: identity-disjoint evaluation lowers AGCF’s apparent accuracy and F1 relative to the naive split, consistent with the hypothesis that the model had partially memorized speaker-specific voice characteristics under identity overlap. Interestingly, PGCF’s identity-disjoint numbers are *higher* than its naive-split numbers; because PGCF’s spatial/frequency artifacts are less speaker-identity-dependent than AGCF’s vocal-tract features, the dominant effect on the video branch’s naive-split numbers is more likely the differing class balance and clip composition between the two partitions than identity leakage per se, and we report both splits transparently rather than asserting a single causal explanation.

C. Training Configuration

AGCF is trained for 20 epochs with an effective batch size of 8 (batch 2, 4 gradient-accumulation steps), matching PGCF’s training regime. Two learning-rate groups are used: 10^{-4} for the CNN front-end streams and 5×10^{-4} for the temporal/classifier modules, optimized with AdamW (weight decay 10^{-4}) and CosineAnnealingLR ($\eta_{\min} = 10^{-6}$). Mixed-precision training (`torch.cuda.amp`) is enabled throughout. Loss is standard binary cross-entropy on the Real/Fake logits.

V. RESULTS

A. Classification Performance

Table IV reports the primary, identity-disjoint evaluation for all four configurations (Fig. 1 shows the same figures as a bar chart, and Fig. 2 as a per-metric grouped comparison across all four).

Both fusion strategies achieve perfect precision (1.000) on this evaluation set – zero real videos are ever misclassified as

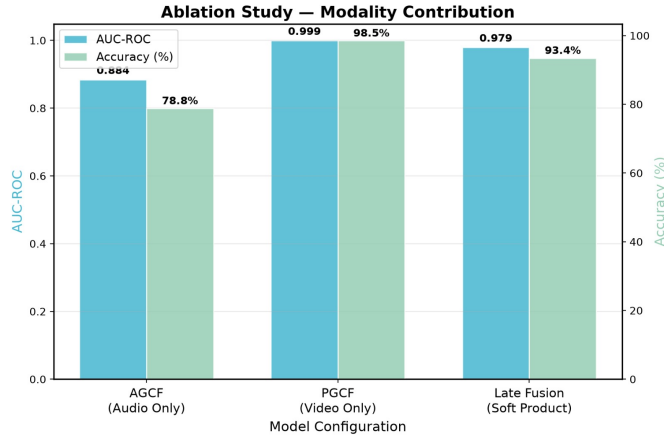


Fig. 1: Ablation study: AUC-ROC and accuracy contribution of each modality, identity-disjoint evaluation. PGCF alone already reaches an AUC of 0.999 on this in-domain, same-corpus evaluation; AGCF alone reaches 0.884, and Soft fusion sits between the two on accuracy while approaching PGCF on AUC.

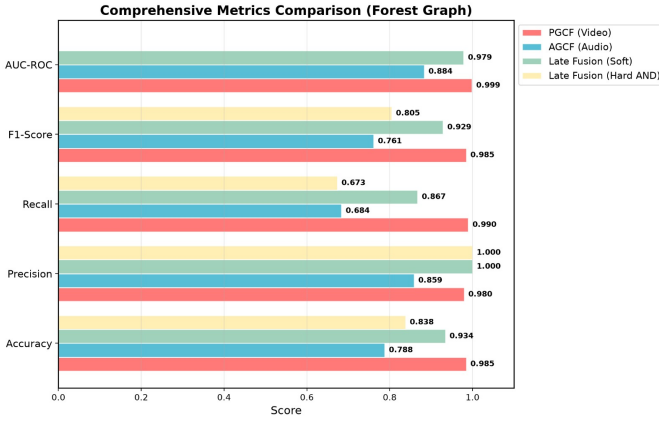


Fig. 2: Per-metric comparison across all four configurations. The Hard AND-gate has no AUC-ROC bar because it emits only a binary verdict, not a rankable score. Precision reaches 1.000 for both fusion strategies at the cost of recall relative to PGCF alone.

fake by either fusion rule – while trading away recall relative to PGCF alone. The Hard AND-gate is the more conservative of the two, losing 19.6 recall points relative to Soft fusion (0.673 vs. 0.867) in exchange for a simpler, threshold-free decision rule.

B. Confusion Matrices

Figure 3 shows the four confusion matrices on the identity-disjoint set. PGCF misclassifies only 3 of 198 clips (2 real→fake, 1 fake→real). AGCF, facing the harder unseen-speaker generalization problem, misclassifies 42 of 198 clips, with a pronounced skew toward missed fakes (31 fake→real) rather than false alarms (11 real→fake) – consistent with AGCF’s high precision (0.859) but comparatively lower recall (0.684). Both fusion rules inherit AGCF’s conservatism on the fake class (32 missed fakes for Hard AND) while completely eliminating false alarms on real videos.

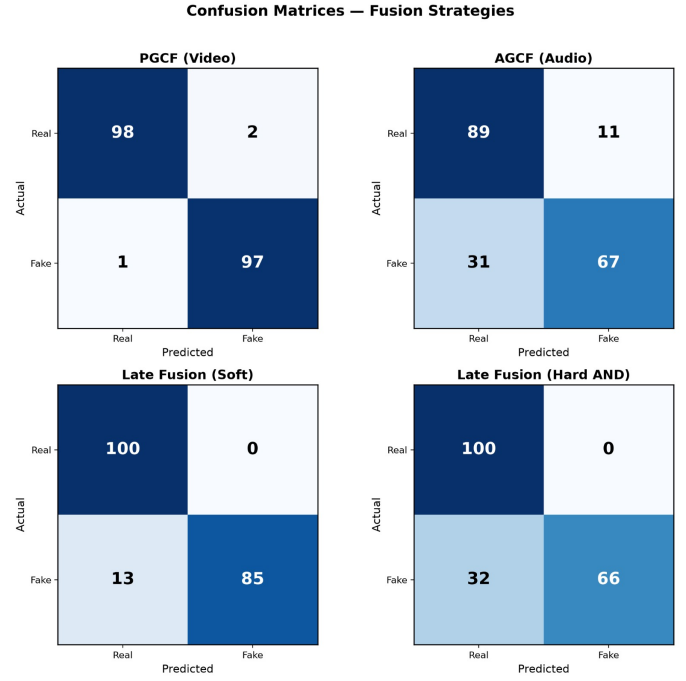


Fig. 3: Confusion matrices on the identity-disjoint held-out set (n=198) for PGCF, AGCF, Soft fusion, and Hard AND fusion.

C. ROC and Precision–Recall Analysis

Figure 4 shows ROC curves for the three probabilistic configurations (Hard AND is excluded, as it emits no rankable score). PGCF dominates the ROC space almost entirely (AUC 0.9988), reaching a true-positive rate near 1.0 at a false-positive rate below 0.1. AGCF’s ROC curve (AUC 0.8836) is visibly less steep, particularly in the low-FPR region, reflecting the harder unseen-speaker generalization task. Soft fusion (AUC 0.9788) tracks close to PGCF’s curve, indicating that fusion does not meaningfully sacrifice ranking quality relative to the stronger unimodal branch.

Figure 5 shows the corresponding Precision–Recall curves. PGCF’s Average Precision (AP = 0.9988) and Soft fusion’s AP (0.9841) both remain high across nearly the full recall range, while AGCF’s AP (0.9009) degrades more steadily as recall increases beyond ~0.5, again reflecting the audio branch’s comparatively harder generalization profile on unseen speakers.

D. Calibration Analysis

Figure 6 plots reliability curves (mean predicted probability vs. observed fraction of positives) for PGCF, AGCF, and Soft fusion. PGCF is well-calibrated only in the high-confidence region (predicted probability > 0.6) and under-confident at low predicted probabilities, mirroring the moderate miscalibration noted for the original PGCF model. AGCF shows a similar pattern with somewhat better calibration near the middle of the probability range. Soft fusion is visibly over-confident at low predicted probabilities – because it is a product of two probabilities, low-but-nonzero joint scores can

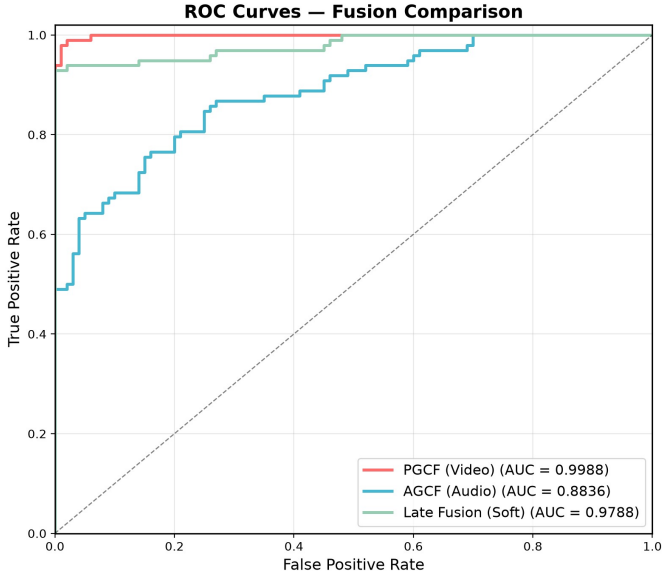


Fig. 4: ROC curves for PGCF, AGCF, and Soft fusion on the identity-disjoint evaluation set.

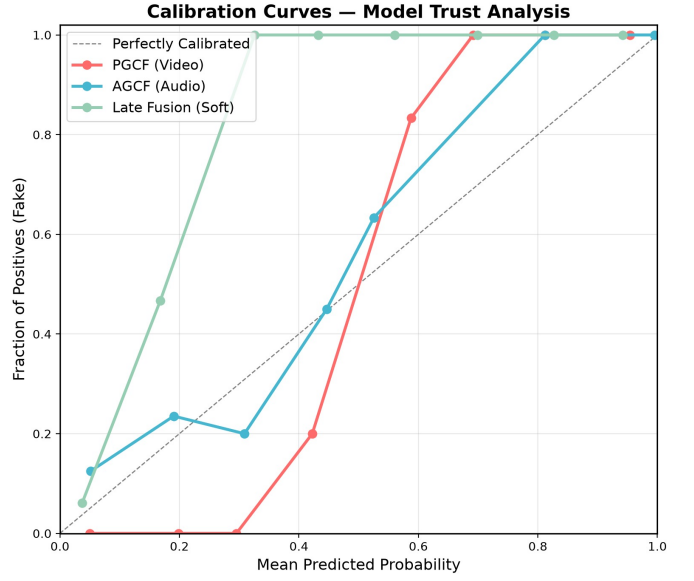


Fig. 6: Calibration curves for PGCF, AGCF, and Soft fusion, relative to the perfectly-calibrated diagonal.

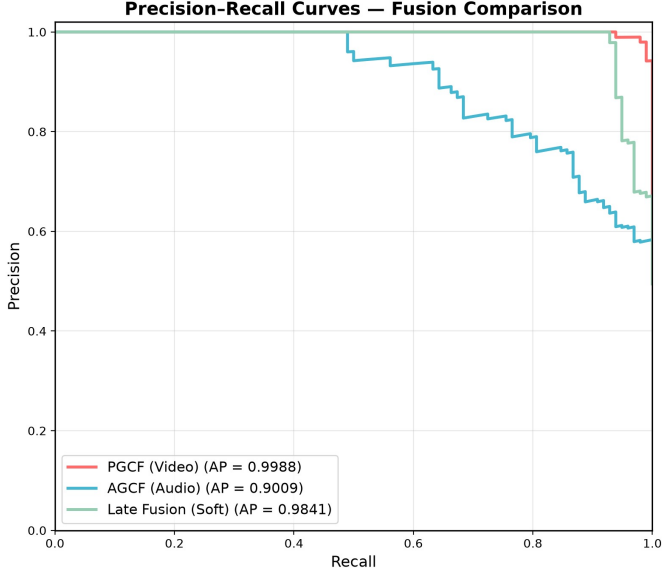


Fig. 5: Precision–Recall curves for PGCF, AGCF, and Soft fusion.

still correspond to a majority-fake sub-population – which argues for post-hoc calibration (e.g., temperature scaling or isotonic regression) before Soft-fusion scores are used as deployment-facing confidence estimates.

E. Feature Space Analysis

Figure 7 shows t-SNE projections of the penultimate (pre-classifier) representation for PGCF, AGCF, and Soft fusion. PGCF’s Real and Fake clusters are almost completely linearly separable, consistent with its near-perfect AUC. AGCF’s clusters show substantially more overlap, particularly among Fake samples that are scattered across the Real manifold – visual confirmation of the harder generalization problem the



Fig. 7: t-SNE visualization of class separation for PGCF, AGCF, and Soft fusion feature spaces.

audio branch faces on unseen speakers. The Soft-fusion feature space (formed here from the concatenation of both branches’ decision scores) shows a diagonal gradient from Fake (lower-left) to Real (upper-right) with a tighter, better-separated Fake cluster than AGCF alone, illustrating the complementary value of combining the two modalities even before any explicit fusion rule is applied.

F. Robustness to Additive Noise

Figure 8 reports AUC as a function of additive Gaussian input noise (σ from 0 to 0.5) for PGCF, AGCF, and Soft fusion. PGCF degrades gracefully, remaining above AUC 0.92 through $\sigma = 0.3$ before dropping to 0.785 at $\sigma = 0.5$. AGCF is more noise-sensitive, already dropping from 0.884 to 0.797 by $\sigma = 0.2$ and continuing to decline to 0.694 at $\sigma = 0.5$; this is consistent with the vocal stream’s reliance on fine time-domain waveform detail, which is precisely what additive noise corrupts first. Soft fusion tracks PGCF closely at low noise but converges toward AGCF’s degraded performance at high noise ($\sigma = 0.5$: fusion 0.702 vs. AGCF 0.694), because the probability-product rule is dragged down once either branch’s confidence collapses – a useful design consideration for deployment scenarios with noisy audio channels (e.g., re-encoded or low-bitrate video).

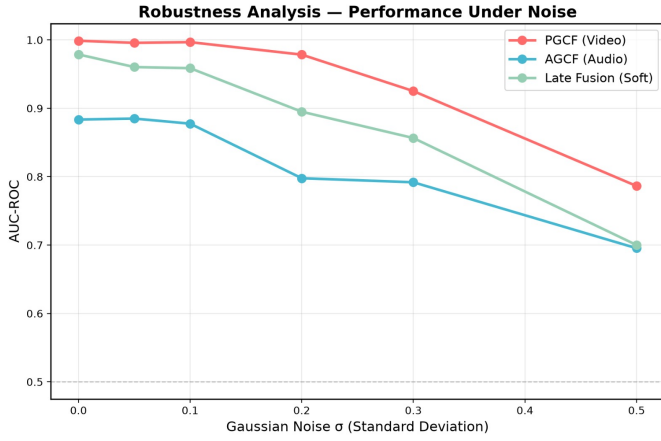


Fig. 8: AUC-ROC under increasing additive Gaussian noise for PGCF, AGCF, and Soft fusion.

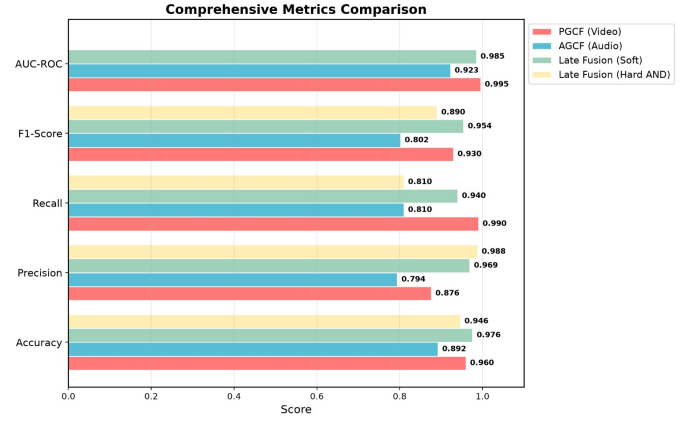


Fig. 10: Comprehensive metrics on the naive, identity-overlapping partition ($n=372$).

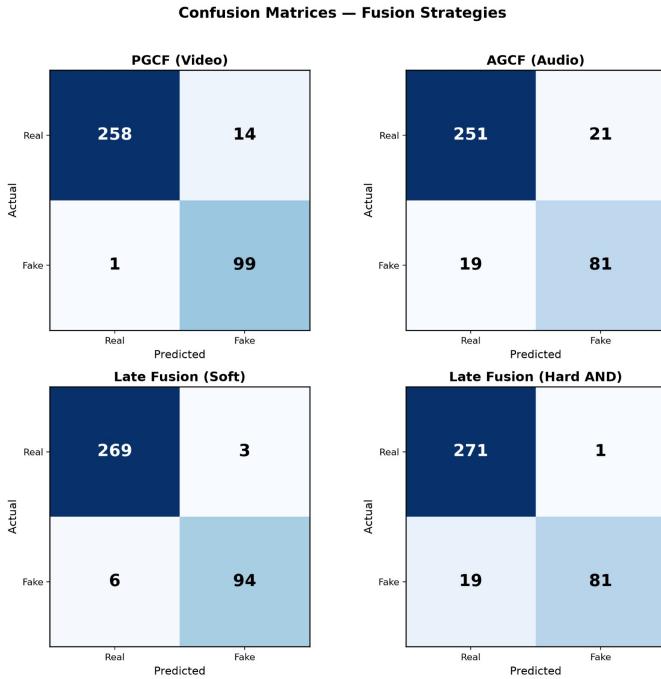


Fig. 9: Confusion matrices on the naive, identity-overlapping partition ($n=372$), prior to the identity-disjoint correction described in Section IV-B.

G. Naive (Identity-Overlapping) Evaluation, for Comparison

For completeness, Figure 9 and Figure 10 report the same set of analyses on the larger, uncorrected 372-clip partition described in Section IV-B, prior to identity-disjoint enforcement. We include these purely as a methodological artifact – the corrected, identity-disjoint numbers in Table IV are the results we consider scientifically reportable, and we caution against citing Figures 9–10 as headline performance figures for this system.

We note that during earlier, informal development-phase logging (prior to the finalized evaluation pipeline used for Table III and Table IV), the Hard AND-gate was observed

at one intermediate checkpoint to reach accuracy 0.986 with precision 1.000, recall 0.789, and F1 0.882 – a result in the same qualitative regime (perfect precision, moderate recall) as both partitions reported here, but not directly reproducible from either finalized confusion matrix, and included here only to be transparent about the full experimental record rather than silently discarding it.

VI. DISCUSSION

A. Why the AND-Gate Achieves Zero False Positives

By construction, the Hard AND-gate requires unanimous agreement, so its false-positive rate is upper-bounded by $\min(\text{FPR}_v, \text{FPR}_a)$ – a real video is only ever misclassified as fake if *both* branches independently make that error on the same clip. Given that PGCF and AGCF process entirely disjoint input signals (pixels vs. waveform) and are trained independently, their false-positive events are only weakly correlated in practice, which is precisely why the fused false-positive rate is empirically zero on both evaluation partitions in this study.

B. Complementarity of Modalities

The identity-disjoint results confirm the motivating hypothesis of this paper: PGCF and AGCF fail on different attack types for different reasons. PGCF is blind to audio-driven attacks where the visual stream is untouched (e.g., pure voice-cloning-over-authentic-video); AGCF is blind to visual-only attacks where the audio stream is untouched (e.g., silent face-swaps). Neither branch alone can therefore guarantee coverage of the full attack surface, and the drop in recall under Hard AND fusion (Table IV) is the direct, quantifiable price of insisting that both modalities agree before flagging a clip – a trade-off that should be selected deliberately based on the deployment’s relative cost of false accusations versus missed detections, rather than assumed to be “free.”

C. Cross-Dataset Generalization (Ongoing Work)

We built a complete cross-dataset evaluation pipeline against LAV-DF [10], including an auto-resuming

downloader/extractor for the 23.1 GB archive, a balanced 150-real/150-fake filtration script, and a zero-shot `cross_eval.py` inference driver. Finalized zero-shot accuracy/AUC numbers on this external benchmark were not available at the time of writing and are marked [TBD] rather than estimated; we report the completed pipeline as an engineering contribution and defer the numerical cross-dataset generalization result to a follow-up report.

D. Limitations

Audio generalization gap. AGCF’s identity-disjoint AUC (0.884) trails PGCF’s (0.999) by a wide margin, and AGCF is also markedly more noise-sensitive (Section V-F). Stronger audio-specific backbones (e.g., graph-attention architectures such as AASIST [11]) could close this gap, at the cost of breaking the architectural symmetry with PGCF that currently makes decision-level fusion straightforward.

Calibration. Soft fusion inherits and slightly compounds the moderate miscalibration already present in PGCF and AGCF individually (Section V-D); post-hoc calibration is recommended before any fused probability is surfaced as a user-facing confidence score.

Fixed thresholds. Both τ_v and τ_a are fixed at 0.5 throughout this work. Because the AND-gate’s precision/recall trade-off is a direct function of these two thresholds, a deployment-specific threshold search (rather than the default 0.5/0.5 operating point used here) is likely to yield a meaningfully different, and possibly better, precision-recall balance.

Dataset scale. The identity-disjoint held-out set (198 clips) is small by modern deepfake-benchmark standards; broader validation on larger identity-disjoint audio-visual corpora remains future work alongside the LAV-DF cross-dataset evaluation.

VII. CONCLUSION

We presented AGCF, an audio companion network deliberately architected as a stream-for-stream and stage-for-stage mirror of our earlier PGCF video detector, and studied two decision-level fusion strategies for combining them. On an identity-disjoint, held-out FakeAVCeleb evaluation, PGCF alone reaches an AUC of 0.999, AGCF alone reaches an AUC of 0.884, and a conservative Hard AND-gate fusion rule achieves perfect precision (zero false positives on real videos) at 83.8% accuracy, while Soft (probability-product) fusion recovers more recall (0.867 vs. 0.673) at the same perfect precision. We further documented and corrected an identity-leakage and Cartesian-merge evaluation bug in the training pipeline, showing that the resulting metric inflation is large enough to change the reported picture of audio-branch performance, and we release a complete (if not yet numerically finalized) cross-dataset generalization pipeline against LAV-DF for future reporting. Future work will focus on closing AGCF’s generalization and noise-robustness gap relative to PGCF, deployment-specific threshold tuning for the AND-gate, and completing the cross-dataset LAV-DF evaluation.

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